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RENEWABLE ENERGY IN NIGERIA: POTENTIALS AND CHALLENGES

尼日利亞的可再生能源：潛力和挑戰

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Abstract

There is a huge renewable potential found in Nigeria which is yet to be explored. The heavy reliance on fossil fuels such as crude oil for its energy production has made it a top contributor to greenhouse gases up to 1.01 % of global emissions. Pertinent efforts have been placed on exploring renewable energy sources given the increased awareness of its sustainability. Furthermore, indicators suggest that crude oil will become a much less significant driver of the energy sector in the coming years. In the light of these issues, this paper presents the potentials that renewable energy has to offer, challenges that could limit its exploitation, and the current status of energy initiatives, including the decade-old renewable energy master plan. Meanwhile, we also discuss the benefits of adopting renewable energy technologies for urban and rural communities. In conclusion, the adverse effect of the exploitation of fossil fuels on the environment, including global warming, glacial meltdown, and acid rain, can be mitigated. Overall, the carbon footprint can be reduced if the potentials are appropriately utilized and converted into an accessible and affordable energy source. Although renewable energy projects have a high initial cost of running over a long life, the cost is small or no compared to conventional sources. The authors recommend the need to explore alternative energy sources that will be environmentally friendly and meet present energy demands.

Keywords: Renewable Energy, Global Emissions, Environment, Sustainable, Carbon Footprint

摘要 在尼日利亞發現了巨大的可再生潛力，尚待開發。能源生產對原油等化石燃料的嚴重依賴使其成為溫室氣體的最大貢獻者，溫室氣體排放量高達全球排放量的 1.01%。鑑於人們對其可持續性的認識不斷提高，已在探索可再生能源方面做出了相關努力。此外，指標表明，未來幾年原油將成為能源部門的重要驅動力。鑑於這些問題，本文介紹了可再生能源必須提供的潛力、可能限制其開發的挑戰，以及包括十年曆史的可再生能源總體規劃在內的能源舉措的現狀。同時，我們還討論了採用可再生能源技術為城市和農村社區帶來的好處。綜上所述，化石燃料的開採對環境帶來的不利影響，包括全球變暖、冰川融化、酸雨等，都可以得到緩解。如果適當利用潛力並將其轉化為可獲取且負擔得起的能源，則可以減少總體碳足跡。儘管可再生能源項目在長期運行過程中的初始成本很高，但與傳統能源相比，成本很小或根本沒有。作者建議需要探索對環境友好並滿足當前能源需求的替代能源。

关键词: 可再生能源、全球排放、環境、可持續發展、碳足跡

I. INTRODUCTION

The renewable energy resources of the Nigerian state have not been fully tapped, resulting in the abundance of use of its non-renewable energy sources such as coal, oil, and gas. These resources are generally termed fossil fuels due to the nature of their formation over the years resulting from the heat and pressure forces acting from within the Earth's core on the relics of dead plants and animals. These energy resources have been explored and exploited in no small measure and usually have drastic environmental consequences. The burning of fossil fuels produces carbon dioxide, contaminating the environs and contributing significantly to global warming. There is also acid rain, rainfall so acidic resulting from atmospheric pollution that it poses significant harm to vegetation and stationary water bodies [1]. The leading cause of acid rain is the emissions of sulfur from power plants and carbon dioxide emissions from petroleum industries [2]. The most explored non-renewable energy in the country is crude oil. It poses a threat to public health and the ecosystem as its exploration activities lead to the release of several toxic compounds, such as benzene, polycyclic aromatic hydrocarbon, etc., into the environment [3]. The aquatic ecosystem is affected by crude oil as its spillage drastically depletes dissolved oxygen in the water needed by aquatic life for survival [4]. When spillage occurs on soil, it results in the organic pollution of groundwater, thereby limiting its usage and causing substantial economic loss, environmental issues, and a decrease in the agricultural productivity of the soil [5]. Crude oil also has the potential for drastic soil nutrient depletion and farmland destruction, and depending on the contamination level, soils polluted by crude oil may remain unsuitable for plant growth for months or years

due to the toxicity of crude oil which persists over a long period [6].

Despite the numerous environmental issues associated with crude oil exploration, it is still a major source of energy in Nigeria as the key to a substantial reduction of poverty for any modern nation is enormous energy use as nations with more significant revenues and higher human development indices are associated with higher energy consumption [7]. Energy is one of the critical drivers of economic development in the 21st century, hence low energy source hurts the different socio-economic growth of a country. An example of such a scenario regarding the agriculture sector is low agricultural production due to a lack of irrigation systems due to inadequate energy supply to power these irrigation systems. Energy in Nigeria primarily refers to electric power; it is a form of energy that can be utilized for various uses due to its flexibility, ease of transmission, and supply [7,8]. Owing to populace outbursts, inevitable industrial development, more agricultural production, and enhancing living standards, the quest for energy in Nigeria is rising.

Given that the country's non-renewable energy resources have been intensely explored, these resources have failed to meet its energy demands. As of May 2015, the current energy generation capacity of the country stood at 3941 MW for a population of over 190 million, with only about 55% of that population having access to electricity [9]. This inadequate energy generation has forced many homes and industries to seek alternatives such as fossil fuel generating plants with their attending noise and atmospheric pollution. It is an unrealistic solution.

The transition from the exploration and utilization of conventional energy-rich resources to renewable resources using appropriate technology in Nigeria is a long-overdue

development that offers a more sustainable approach to rural development, healthy ecosystem, energy independence, and climate change mitigation [8]. The International Energy Agency (IEA) defines renewable energy as the energy that is gotten from natural processes that are regenerated at a faster rate than they are consumed [10]. It is necessary to point out that energy being renewable does not necessarily imply sustainability. The sustainability approach integrated into the use of renewable energy would also ensure that its generation and collection, conversion and transmission, and utilization occur in a way that does not contribute adversely in a negative way to the environment and its natural ecosystems [8]. Renewable energy sources include hydropower, solar energy, biogas, biomass, geothermal, wind, and water waves; they have numerous applications in electricity, fuels, thermal energy, and mechanical force. The two significant advantage of renewable energy is environmental friendliness and energy security. Renewable energy has little or no effect on the environment. It produces very little waste while also ensuring the by-products like carbon dioxide, sulfur dioxide, and other chemicals will be minimized. Renewable energy, according to environmental analysts, reduces greenhouse gas emissions by at least 3.2 kilograms of carbon dioxide per kilogram of biodiesel, a reduction of about 99% for emission of sulfur dioxide, 20% for carbon, 32% for carbon dioxide, 50% for soot and 39% for particulate matter reduction [11]. Renewable energy also improves energy security by providing greater diversity in the overall energy mix and reducing dependency on the non-renewable energy source; It also offers affordable rural energy without access to a centralized energy system [11].

II. CLIMATE CHANGE IN NIGERIA

Coster et al. [12] describe climate change as the alteration in the global climate system that leads to new, long-term weather patterns and global warming as the extended increase in Earth's average climate system temperature. The greenhouse effect leads to a global warming process, a natural process whereby the atmosphere maintains particular heat from the Sun, and the Earth maintains conditions necessary for its life. The adverse effect of the exploration and exploitation of fossil fuels on the environment has created significant problems. The combustion of these fuels has increased greenhouse gases present in the atmosphere; this is the principal cause of heightened climate change. There are quite a several diverse impacts

that climate change will have on sustainability. The identified evidence includes but is not restricted to the thermal expansion and meltdown of glaciers on land. In its turn, it led to an increase in sea levels resulting in submerged land and habitat loss, the devastation of farmlands, saltwater intrusion, and so on. The planet's ecosystem is at serious threat owing to global warming, and the coastal regions are the most vulnerable to this phenomenon. With a large portion of Nigeria's urban population situated in these regions, there is an urgent need to mitigate this problem. The negative effect of climate change is being experienced in the country, with environmental issues such as persistent droughts and flooding affecting the welfare of millions of people [12, 13]. Our natural resources, cultural assets, and built infrastructure take some of the hit with yearly flood episodes, which have displaced citizens and destroyed properties. The alteration of rainfall regimes and the hydrological cycle sequence due to climate change and global warming would affect agricultural expanses; rise in relative humidity and temperature provides a hub for diseases and pathogens to thrive; climate change would also affect the tourism sector revenue. A defining component in mitigating climate change and global warming is to achieve minimum anthropogenic release of carbon dioxide into the environment, which is far-reaching and represents the most significant part of greenhouse gas emissions. As one of Africa's largest greenhouse gas emissions producers, Nigeria contributes to global warming via carbon dioxide produced in the Niger Delta region due to continuous gas flaring [14]. The vast crude oil reserve of the country is located in this region, with over 11 companies exploiting this resource which has resulted in flaring up of gases of up to 17 billion cubic meters, including methane (about 12 million tons), carbon dioxide (3.5 million tons) and sulfur dioxide (160 tons) are released into the atmosphere [2]. As a country, Nigeria depends heavily on fossil fuels, and due to the environmental damage associated with fossil energy exploitation and exploration, there is a need to explore other environmentally friendly energy sources to meet our energy demands [2]. This necessity can be met by increasing renewable energies, including hydro, biogas, wind, and solar resources, and adopting sustainable urban energy planning methods [2].

III. RENEWABLE ENERGY SOURCES

Over the years, a relevant effort has been placed on exploring renewable energy sources, given the increased awareness of its sustainability.

Conversely, there are indicators that crude oil will become not as much of a significant driver in the energy sector as it is currently and has necessitated exploring renewable energy resources in Africa. Despite the epileptic condition of the energy sector in terms of generation and distribution in the country, Nigeria has a wealth of renewable energy resources, including large and small hydroelectric dams, solar, biomass, wind energy, and geothermal energy. This chapter further discusses the four major ones concerning power generation.

A. Solar Energy

Solar energy can be defined as harnessed from sunlight for the generation of power. It depends on the nuclear fusion power from the Sun's core. There are two methods by which we can obtain energy from the Sun; direct and indirect methods. The natural procedure involves photovoltaic (PV), while the indirect way is to use sunlight to boil and heat water to supply power. The latter method is referred to as the concentration of solar energy (CSP). The generation of power through the use of the Sun's radiation is principally regarded as solar power. Still, except for geothermal energy and tidal energy, all renewable energy sources have the Sun as their primary source. Nigeria has a landmass of about 924 km² with an average irradiance per unit area of 5.535 KWh/m² and an average incident solar energy of 1831.06 kWh annually [15]. The country's sunshine hours range from 9 hours in the far north to 3.5 hours in the southern part of the country [16]. With an average of 6 hours per day of sunlight, the Government receives about 4.85×10^{12} kW/h of the Sun's energy (3.8×10^{23} kW/s) per day, equivalent to the energy derived from about 1.082 million tons of oil per day [15].

These statistics show that a considerable amount of energy can be generated and utilized from the massive solar radiation available to combat the epileptic energy supply, leading to an absence of economic development [16]. There are various ways of exploiting solar energy in Nigeria, including solar photovoltaic cells for rural electrification; Cookers, crop dryers, water pumps, and heaters, amongst others, can harness power from others the Sun. Solar energy utilization is on the rise in Nigeria, especially in street-lighting, water-pumping, rural electrification, refrigeration, showering of communication stations, etc. [17]. For example, a fully operational solar PV manufacturing plant has been set up in Abuja by the National Agency for Science and Engineering Infrastructure

(NASENI), with a current capacity of 7.5 MW of power generation per year. It plans to upgrade the plant capacity [18]. In Delta state, the Government signed a Memorandum of Understanding with Yutal Li Nigeria Limited in 2016 to construct a 100 MW solar power plant to promote clean energy and the growth of micro, small, medium scale enterprises [19].

B. Wind Energy

The exploitation of wind as a means of energy provision for power generation for valuable purposes is termed wind energy. It depends on the temperature difference at the Earth's surface. It is often the case because the character of the earth changes at different temperatures as the Sun shines [7]. Typical wind power devices include electricity-generating wind turbines and windmills and wind and drainage pumps used for water abstraction. The wind strength changes concerning location; hence total energy efficiency cannot be entirely operated when considering only the average wind strength of a wind turbine at that location. By nature, wind energy is clean, abundant, affordable, inexhaustible, and environmentally friendly, and these advantages have made it one of the fastest growth rates in the renewable energy market worldwide [20]. The cumulative installed capacity of wind power in the world has increased progressively in the past few years from 7600 MW in 1997 to 318,117 MW in 2013, with the value in 2013 higher than the 263,126 MW in 2011, 198,001 MW in 2010, and 158,975 MW in 2009 [21].

In Nigeria, wind energy has potential, with most of Nigeria having an average wind speed of 3.0 m/s with a wind speed that ranges from 3.88 m/s to 9.39 m/s in the North-West part of the country, 1.77 m/s to 4.5 m/s in the South-West, 2.46 m/s to 5.36 m/s in the North-Central, 3.30 m/s to 4.65 m/s in the South-South [22]. Despite the viability of wind energy in Nigeria, the generation of power from wind is still developing in the country as no wind farms are contributing to the national grid [23]. Small-scale wind turbines have been designed and used for purposes in the country such as water pumping, battery charging, small-scale enterprise electrification, and experiments. The only large-scale onshore wind turbine in the country is located in the Rimi Local Government Area of Katsina state, with an estimated output capacity of 10 MW [24]. Small-scale wind turbines that exist in the country include the 5 kW wind energy system installed in Sokoto (Sayyan Gidan Gada) and the 1 kW of wind power turbines

installed in Bauchi (Kedada) and Katsina (Goronyo) for pumping water [25]. The tremendous potentials and generation capabilities that could be harnessed from wind energy have not been fully utilized, and gross energy yield is in its initial stage. According to [14], the country is yet to explore its wind energy resource, improving the standard of living for its citizens, especially those in the rural regions.

C. Hydro Energy

Hydro energy is the utilization of falling or flowing water's gravitational force to generate power. It is the most common form of renewable energy found in nearly every corner of the planet. It relies on high water's gravitational potential as lifted by sunlight from the oceans [7]. Hydropower plants are usually situated in big dams with high gravitational forces. Their power generation does not produce any direct or indirect by-products. It is regarded as an energy source with a lower level of greenhouse gas output. This power source accounts for approximately 20% of the world's electricity and 88% of the renewable energy generated. Hydro energy is subject to seasonal droughts as its capability is subject to annual rainfall levels. Its distribution and the volume of water coming from the upstream end of the river. In Nigeria, hydropower is possibly the most utilized power resource for electric power generation and supply as the country's wealth of large rivers and waterfalls has been adequately leveraged. The country's total hydropower is 14,750 MW with 11,250 MW in Large Hydropower and Small Hydro with a 3,500 MW share. The potential of Hydropower energy in Nigeria is high and accounts for well over 38.5% of the country's total electricity power supply between 1999 and 2004. Yet only about 1930 MW, which constitutes 14% of the total exploitable hydropower in the country, has been utilized, contributing over 30% to the installed capacity connected to the grid [26]. The country has two major hydropower stations, with one located at the Kainji Dam with an installed capacity of 160MW and the other at the Jebba dam approximately 82 km downstream of the former on the River Niger installed capacity of 96.4 MW [27]. Other rivers in the country with energy generation potential include rivers Kaduna, Benue, and Cross River, which could produce a total capacity of about 4650 MW.

In comparison, the river Mambilla can also generate about 2330 MW [28]. Though the country uses major rivers for hydro energy generation, there is the potential to utilize smaller rivers as small hydro schemes to meet energy

demands of various rural communities either as a single electric power source or in combination with other renewable energy sources as a hybrid electric power source [29-31]. Small hydropower can contribute immensely to global energy, especially in developing countries like Nigeria [32]. The inherent advantages of small hydropower systems, including minimal civil works, the low magnitude on the environment, non-excessive power generating technology, and the potential for flood prevention, navigation, irrigation, and fisheries, made them rapidly popular in developing and developed economies around the world. The SHPs development can offer a sustainable means for meeting the needs of the rural dwellers while improving the standard of living in the community, transforming them into urban centers.

D. Biomass Energy

Biomass energy is the generation of power from biological materials, substances, or organisms such as waste, wood, and alcohol fuels [33]. Biomass such as dead trees and wood chips can be used to generate electricity and heat. Very often, this waste is converted to fuel during combustion. Biomass as energy sources do not include organic fossil fuels; however, they occur in various forms and can be used as fuel for automobiles. Electric power plants are modern forms of biomass energy. Several methods for transforming biomass into electricity are available: the most common practice is direct combustion of biomass material like farm waste or wood waste. Gasification, pyrolysis, and anaerobic digestion are other approaches. Gasification is a technique that converts biomass or fossil fuel-based materials into synthesized gas with usable energy content. This process usually involves less Oxygen content than is required for the total combustion of the biomass. In the absence of oxygen, pyrolysis produces bio-oil by rapidly heating the biomass. The digestion by anaerobic bacteria (decomposition) of organic matter would yield renewable natural gas without oxygen. Biomass feedstocks can be transformed into different fuels through a range of heat or electricity generation processes and produce liquid biofuels or biogas [34].

Nigeria is rich in biomass resources, including wood, forage materials, and shrubs, including agricultural waste and by-products from municipal and industrial actions.

Oyedepo [34] estimated the overall biomass and bioenergy potentials in Nigeria to be about 200 billion kg/year and 2.58 billion GJ (61.67

Mtoe), respectively, representing about 51% of total energy consumption in the country in 2015.

With an energy content of 6.0×10^9 MJ, Nigeria exploits 80 million m^3 (4791.7kg) of fuelwood yearly for cooking, amongst other domestic purposes. Out of this energy content, only about 5% to 12% are currently used for domestic purposes especially cooking. The entire primary fuelwood and charcoal energy consumption were put at 32-40% between 1989 and 2000, and the country's national demand was estimated at 39 million tons for the year 2000. Household cooking and cassava processing industries were responsible for 95% of total fuelwood consumption.

The megawatt-biomass gas turbine project in Ebonyi State is a modern example of biomass exploitation in the country that utilizes the husks of plants to generate energy in its contribution to the fast development of the state. The five megawatts to be developed will provide energy of about two megawatts to the mills, and the Oferekepe waters will use 1.5 megawatts with the balance going to adjoining communities [30]. Power pallets, pioneering technology for making renewable energy from biomass (waste), have been introduced. These machines look like small generators and are available in 10kW/25KVA and 20kW/38 KVA sizes, with the ability to generate power through the conversion of woody biomass to electricity and heat [30]. Over the years, biomass availability in the country has been drastically cut down, partly due to the demand for timber in the construction and furniture industries. An estimated 200 million tons of dry biomass is obtainable from forage, grasses, and shrubs, producing a total of 2.28×10^6 MJ of energy [33]. Most people living in rural regions depend mainly on wood for shelter and fuel for burning, leading up to an annual loss of approximately 350,000 hectares of forest and vegetation by far lower afforestation rate of 50,000 hectares per year [34]. This act involving the continuous felling of the tree will eventually lead to desert encroachment and soil erosion if left unchecked. To address this problem, the introduction of affordable and acceptable solar stoves needs to discourage fuelwood use. It will reduce the consumption of wood and allow for better combustion. Other merits include reducing the cooking time and provision of an organized channel for the smoke exit. The reduction in the area of land by sea levels and floods in the South, and droughts in the north is another reason behind the decrease in the biomass availability of the country. Biomass is undoubtedly a significant renewable energy source, but it must be clearly

understood in sustainable production. Nigeria should use its oil palms, woodland, municipal waste, rice, and sugar cane husk for biogas production with its abundant biomass resources. Sugar factories can use their cane residues and waste in their country. In contrast, paper mills and packaging factories can use waste biomass in South Africa and Malaysia to generate process steam. This energy can significantly reduce greenhouse gas emissions when converted to liquid fuel for cooking instead of wood for fuel.

IV. RENEWABLE ENERGY MASTER PLAN

Nigeria's quest to promote the diversification of its energy mix gave birth to the Renewable Energy Master Plan (REMP), approved by the Government in November 2005 and later updated in 2011. The Energy Commission of Nigeria (ECN) and the United Nations Development Program (UNDP) worked on this plan to establish a general vision, goals, and road map to resolve key energy issues the country faces through speedy development and use of renewable energy. REMP sets vivid and verifiable National targets for short-term, medium, and long-term use by an inclusive framework for renewable energy policy development, legislation, technology, human resources, infrastructure, and the market. Short-term targets were set for 2007, medium-term targets for 2015, and long-term targets set for 2025. The implementation of the REMP must take account of the country's sustainable development objectives, as set out in its National Development Policy documents known as Nigeria Vision 2020 and the NASPA-CCN, National Adaptation Strategy, and Climate Change Action Plan for Nigeria [2, 35]. It is anticipated that the plan will facilitate the transition of energy from crude oil to a less carbon-intensive economy and increase the need for renewable energy. More than a decade has passed since the beginning of the REMP. No specific progress has yet been seen in this field; aside from existing hydroelectric power stations, no other additional grid generation of electricity from renewable sources has been documented [2].

In 2007 the Federal Executive Board approved the NNPC's (Nigerian National Petroleum Corporation) Bio market Policy Framework, a new plan to build renewable energy technology in Nigeria. The Nigerian Government has explored large-scale biofuels production for application in automobiles, generators, and bio-gas production. A 70,000-euro grant was given to the NNPC to fund a

comprehensive feasibility study by the Renewable Energy and Energy Efficiency Partnership (REEEP) in support of a comprehensive feasibility study for the conversion of high-ethanol yielding cassava varieties for the production of two types of automotive fuel: ethanol fuel and palm oil diesel fuel [36]. However, renewable energy and alternative forms such as solar photovoltaic, wind, hydroelectric, and energy efficiency are relatively recent. Unfortunately, the current work in the country on renewables' is limited and mainly anchored by a foreign and non-governmental organization, particularly in remote rural areas. Furthermore, in big cities in the country, the erratic power supply in the national grid at Oshogbo is hardly standalone in installing renewable energies.

V. PROBLEMS OF RENEWABLE ENERGY EXPLOITATION IN NIGERIA

Several authors have highlighted and discussed the numerous challenges and obstacles hindering the exploitation of renewable energy in Nigeria. Authors in [2] stated that the major hindrance to renewable energy exploitation is the nonexistence of applicable laws through which sustained implementation of the renewable energy plan in the country can be achieved considering that the previous master plan (REMP) is a little more than a mere policy with no legal backing. The non-availability of funds to carry out projects and the rather poor incentives accompanying investments have brought a halt to renewable energy projects and contributed significantly to the failure of the REMP. The various barriers to the success of the REMP, as discussed by [8], are policy and political barriers, market distortions, international development barriers, standards quality and control barriers, research and development barriers, and environmental barriers. In addition, appropriate laws will ensure that technologically advanced nations do not turn the country into a dumping ground with sub-standard goods. These laws may also impose zero taxes on renewable products to benefit the poorest of the poor, which represents the majority of the Nigerian population [30]. Another key obstacle to the exploitation of renewable energy in Nigeria is technical capability. The choice of materials, designs, and brand type of the turbines are tailored to be site-specific. Oyedepo et al. [37] showed that the major technical challenge with wind turbines would be the dependence on manufacturers to

design, fabricate, and maintain the turbines. At the same time, this provision is not available in Nigeria. It creates significant gaps in the production and maintenance capacity of system components. Furthermore, this challenge is heightened by the location of these projects in remote areas, making it difficult to attract a proficient and experienced workforce for operations. Adejumobi and Shobayo [29] discussed technical issues as well, stating that average Nigerians have the insufficient technical know-how, in particular in areas where there are insufficient and incorrect data on resources, low quality of products, little or no research and development work, and lack of human and manufacturing capabilities, to maintain modern trends in renewable energy technologies. Nyika et al. [30] stated that the lack of insight into local energy needs, research, and development has also not been established to carry out reliable and effective designs that can meet local demands. The lack of local skilled labor to properly install, run or maintain equipment, lack access to replacement parts, and appropriate public awareness is another significant obstacle to implementing renewable energy. The lack of adequate awareness by the public of the potential for renewable electricity in dealing with some of the country's challenges for energy and development distorted the market, which led to a higher perception that potential renewable energy projects are unreliable, and as such, suited for just niche markets [37],[38]. As stated by [39], the development, application, and dissemination of renewable energy resources and technologies in the Nigerian energy market have been restricted due to the lack of awareness of these resources' economic and environmental benefits. Akinbami [40] added that this low level of awareness had handicapped the public, making it difficult to influence the Government into taking more vital initiatives towards improving the growth, application, distribution, and circulation of renewable energy resources and technologies sustainable national energy market. It can be attributed to the general public being embroiled in the illusion that the country has an unlimited source of fossil fuels (oil, natural gas, coal). Nyika et al. [30] noted that economic and financial barriers are another major hindrance to renewable energy systems development in Nigeria. Challenges arise from lack of access to capital to lack of life support, and the lack of investment to lack of information supply by appropriate financial institutions. Others include the scale of energy systems, inappropriate government subsidies, or other agencies.

Renewable electricity projects incur high initial and are rare. It makes it difficult to obtain loans, with bankers attributing a high degree of risk to such projects and giving preference to large-scale conventional electricity investments [37]. Even when such loans are available, the interest rates are generally high, which discourages many investors. Okedu et al. [38] pointed out that although renewable energy projects have a high initial cost of running over a long life, the cost is small or no compared to conventional sources like diesel. Still, most low-income consumers prefer to decrease initial costs rather than decrease operating costs. Nyika et al. [30] concluded that renewable energy systems costs in Africa might remain high due to the high financial input and low-profit margins during manufacturing of the component elements due to the low sponsorship and high R&D costs. Other notable challenges or obstacles to the exploitation of renewable energy in Nigeria include Social, Cultural, Religious, and environmental constraints.

VI. CONCLUSION

Nigeria has a wealth of renewable energy resources. Despite the numerous environmental issues associated with conventional energy resources and their inability to satisfy the country's energy demands, these renewable energy resources are yet to be fully explored to meet their vast potentials. It poses a considerable problem to agriculture as various agricultural activities depend on energy supply for efficient agricultural production. Renewable energy has a crucial role in mitigating climate change. This phenomenon is primarily caused by the emission of greenhouse gases associated with the exploitation of conventional energy sources such as crude oil.

The country's potential for renewable resources includes large and small hydropower, solar, biomass, wind, hydrogen use potential, and geothermal and ocean energy development. Despite efforts such as the Renewable Energy Master plan, several challenges have made the exploitation of these energy resources difficult. These challenges should be addressed to fast-track the exploitation of renewable energy in the country, solve the country's epileptic power situation, and aid climate change mitigation.

REFERENCES

- [1] ABALI, H.W., ETEBU, O.M. and LETON, T.G. (2018) Corossitivity of pollutants from gas flares on galvanized roofing sheets in Idu, Rivers State, Niger Delta of Nigeria. *Journal of Environment Pollution and Human Health*, 6(1), pp.1-6.
- [2] ELUM, Z.A. and MOMODU, A.S. (2017) Climate change mitigation and renewable energy for sustainable development in Nigeria: A discourse approach. *Renewable and Sustainable Energy Reviews*, 76, pp.72-80.
- [3] NGENE, S., TOTA-MAHARAJ, K., EKE, P., and HILLS, C. (2016) Environmental and economic impacts of crude oil and natural gas environmental and economic impacts of crude oil and natural gas production in developing countries. *International Journal of Economy, Energy, and Environment*, 1, pp.64-73.
- [4] CHIKWE, T.N., and OGBOLE, S.N. (2019) Remediation of simulated oil-contaminated sites using shells of clams and oyster-total petroleum hydrocarbons of simulated oil-contaminated sites, before and after remediation. *Nigerian Journal of Chemical Research*, 24(1), pp.57-67.
- [5] OBIKALAIJE, U.M., MAKINDE, O.A. and AMAKOROMO, E.R. (2015) Bioremediation of crude oil polluted soil using animal waste. *International Journal of Environmental Bioremediation & Biodegradation*, 3(3), pp.79-85.
- [6] BENSON, D.M., OCHEKWU, E.B. and TANEE, F.G. (2016) Enhancement of crude oil polluted soil by applying single and combined cow-dung and hydrogen peroxide as remediating agents. *Journal of Applied Sciences and Environmental Management*, 20(4), pp.1137-1145.
- [7] AKURU, U.B., ONUKWUBE, I.E., OKORO, O.I. and OBE, E.S. (2017) Towards 100% renewable energy in Nigeria. *Renewable and Sustainable Energy Reviews*, 71, pp.943-953.
- [8] ENERGYPEDIA. (2019, May 27). Nigeria Energy Situation. [Online] Available from: <https://www.energypedia.info/wiki/NigeriaEnergySituation>. [Accessed 18/07/21].
- [9] AKURU, U.B. and OKORO, O.I. (2014) Renewable energy investment in Nigeria: A review of the Renewable Energy Master Plan. *Journal of Energy in Southern Africa*, 25(3), pp.62-67.

- [10] HARJANNE, A. and KORHONEN, J.M. (2019) Abandoning the concept of renewable energy. *Energy Policy*, 127, pp.330-340.
- [11] CHILAKPU, K.O. (2015) Renewable energy sources. Their benefits, potentials, and challenges in Nigeria. *Journal of Energy Technologies and Policy*, 5(9), pp.21-24.
- [12] COSTER, A.S. and ADEOTI, A.I. (2015) Economic effects of climate change on maize production and farmers' adaptation strategies in Nigeria: A Ricardian approach. *Journal of Agricultural Science*, 7(5), p.67.
- [13] WIKIPEDIA. Climate change. [Online] Available from: https://en.wikipedia.org/w/index.php?title=Climate_change&oldid=1026669630. [Accessed 18/07/21].
- [14] SHAABAN, M. and PETINRIN, J.O. (2014) Renewable energy potentials in Nigeria: Meeting rural energy needs. *Renewable and Sustainable Energy Reviews*, 29, pp.72-84.
- [15] AGBO, E.P., EDET, C.O., MAGU, T.O., NJOK, A.O., EKPO, C.M. and LOUIS, H. (2021) Solar energy: A panacea for the electricity generation crisis in Nigeria. *Heliyon*, 7(5), p.e07016.
- [16] ABDULLAHI, D., SURESH, S., RENUKAPPA, S. and OLOKE, D. (2017) August. Key barriers to the implementation of solar energy in Nigeria: a critical analysis. *2017 IOP Conference Series: Earth and Environmental Science*, 83(1), 012015.
- [17] ADEKUNLE, O.R., JAAFAR, A.B. and MUSA, M.N., (2015) The potentials of solar thermal technology for sustainable development in Nigeria. *International Journal of Engineering and Technical Research*, 3(7), pp.2454-4698.
- [18] OZOEGWU, C.G., MGBEMENE, C.A. and OZOR, P.A. (2017) The status of solar energy integration and policy in Nigeria. *Renewable and sustainable energy reviews*, 70, pp.457-471.
- [19] AUDU, M.O., TERWASE, A.S. and ISIKWUE, B.C. (2019) Investigation of wind speed characteristics and energy potential in Makurdi, north-central, Nigeria. *SN Applied Sciences*, 1(2), 178 (2019).
- [20] ARIRIGUZO, J.C., and EKWE, E.B. (2019) Weibull distribution analysis of wind energy prospect for Umudike, Nigeria for power generation. *Robotics and Computer-Integrated Manufacturing*, 55, pp.160-163.
- [21] AKOREDE, M. F., O. IBRAHIM, S. A. AMUDA, A. O. OTUOZE, and OLUFEAGBA, B. J. (2017) Current status and outlook of renewable energy development in Nigeria. *Nigerian Journal of Technology*, 36(1), pp.196-212.
- [22] OLULEYE, A. and ADEYEWA, D. (2016) Wind energy density in Nigeria as estimated from the ERA-interim reanalyzed data set. *Current Journal of Applied Science and Technology*, 17(1), pp.1-17.
- [23] ADEDIPE, O., ABOLARIN, M.S. and MAMMAN, R.O. (2018) A review of onshore and offshore wind energy potential in Nigeria. *2018 IOP Conference Series: Materials Science and Engineering*, 413(1), 012039.
- [24] MOHAMMED, Y.S., MUSTAFA, M.W., BASHIR, N. and IBRAHIM, I.S. (2017) Existing and recommended renewable and sustainable energy development in Nigeria based on autonomous energy and microgrid technologies. *Renewable and Sustainable Energy Reviews*, 75, pp.820-838.
- [25] EBHOTA, W.S., and TABAKOV, P.Y. (2018) The place of small hydropower electrification scheme in socio-economic stimulation of Nigeria. *International Journal of Low-Carbon Technologies*, 13(4), pp.311-319.
- [26] OMEIZA, I.O.A., OJONGBEDE, H.A., THOMAS, C.T. and OLUFEAGBA, B.J. (2019) Assessment of performance of turbo-alternators at the Jebba Hydroelectric Power Station in Nigeria from 2005–2014. *Nigerian Journal of Technology*, 38(1), pp.134-141.
- [27] OKEDU, K.E., UHUNMWANGHO, R. and ODJE, M. (2020) Harnessing the potential of small hydropower in the Cross River state of Southern Nigeria. *Sustainable Energy Technologies and Assessments*, 37, 100617.
- [28] OGUEKE, N.V., IKPAMEZIE, I.I. and ANYANWU, E.E. (2016) The potential of a small hydro/photovoltaic hybrid system for electricity generation in FUTO,

Nigeria. *International Journal of Ambient Energy*, 37(3), pp.256-265.

[29] ADEJUMOBI, I.A., and SHOBAYO, D.I. (2015) Optimal selection of hydraulic turbines for small hydroelectric power generation – A case study of Opeki River, southwestern Nigeria. *Nigerian Journal of Technology*, 34(3), pp.530-537.

[30] NYIKA, J., ADEDIRAN, A.A., OLAYANJU, A., ADESINA, O.S., and EDOZIUNO, F.O. (2020) The potential of biomass in Africa and the debate on its carbon neutrality. [Online] Available from: <https://www.intechopen.com/online-first/the-potential-of-biomass-in-africa-and-the-debate-on-its-carbon-neutrality>. [Accessed 18/07/21].

[31] BEN-IWO, J., MANOVIC, V., and LONGHURST, P. (2016) Biomass resources and biofuels potential to produce transportation fuels in Nigeria. *Renewable and Sustainable Energy Reviews*, 63, pp. 172-192.

[32] OLANREWAJU, F.O., ANDREWS, G.E., LI, H. and PHYLAKTU, H.N. (2019) Bioenergy potential in Nigeria. *Chemical Engineering Transactions*, 74, pp.61-66.

[33] IGBINOVIA, F.O. (2014) An Overview of renewable energy potentials in Nigeria: prospects challenges and the way forward. *Energetika Journal*, 46, pp.570-579.

[34] OYEDEPO, S.O. (2012) On energy for sustainable development in Nigeria. *Renewable and sustainable energy reviews*, 16(5), pp.2583-2598.

[35] SAMBO, A.S. (2009) Strategic developments in renewable energy in Nigeria. *International Association for Energy Economics*, 16(3), pp.15-19.

[36] DIJI, C.J. (2013) A Critical review of the Nigerian Energy Scenario. *IOSR Journal of Electrical and Electronics Engineering*, 5(3), pp.55-67.

[37] OYEDEPO, S.O., BABALOLA, P.O., NWANYA, S., KILANKO, O.O., LERAMO, R.O., AWORINDE, A.K., ADEKEYE, T., OYEBANJI, J.A., ABIDAKUN, O.A. and AGBEREGHA, O.L. (2018) Towards a sustainable electricity supply in Nigeria: the role of the decentralized renewable energy system. *European Journal of Sustainable*

Development Research, 2(4), Article No.40. [Online] Available from: <https://www.ejosdr.com/article/towards-a-sustainable-electricity-supply-in-nigeria-the-role-of-decentralized-renewable-energy-3908>.

[38] OKEDU, K.E., UHUNMWANGHO, R. and WOPARA, P. (2015) Renewable energy in Nigeria: The challenges and opportunities in mountainous and riverine regions. *International Journal of Renewable Energy Research (IJRER)*, 5(1), pp.222-229.

[39] NNAJI, C.E., UZOMA, C.C. and CHUKWU, J.O. (2010) The role of renewable energy resources in poverty alleviation and sustainable development in Nigeria. *Continental Journal of Social Sciences*, 3(31), pp.31-37.

[40] AKINBAMI, J.F.K. (2001) Renewable energy resources and technologies in Nigeria: present situation, prospects, and policy framework. *Mitigation and adaptation strategies for global change*, 6(2), pp.155-182.

參考文:

- [1] ABALI, H.W., ETEBU, O.M. 和萊頓, T.G. (2018) 尼日利亞尼日爾三角洲伊杜河州鍍鋅屋面板上的氣體燃燒污染物的腐蝕性。環境污染與人類健康雜誌, 6(1), 第 1-6 頁。
- [2] ELUM, Z.A. 和 MOMODU, A.S. (2017) 氣候變化減緩和可再生能源促進尼日利亞可持續發展：一種話語方法。可再生和可持續能源評論, 76, 第 72-80 頁。
- [3] NGENE, S., TOTA-MAHARAJ, K., EKE, P. 和 HILLS, C. (2016) 原油和天然氣的環境和經濟影響發展中國家。國際經濟、能源和環境雜誌, 1, 第 64-73 頁。
- [4] CHIKWE, T.N. 和 OGBOLE, S.N. (2019) 使用蛤殼和模擬石油污染場地的牡蠣總石油烴修復模擬石油污染場地, 修復前後。尼日利亞化學研究雜誌, 24(1), 第 57-67 頁。
- [5] OBIKALAIJE, U.M., MAKINDE, O.A. 和 AMAKOROMO, E.R. (2015) 使用動物糞便對原油污染土壤進行生物修復。國際

環境生物修復與生物降解雜誌, 3(3), 第 79-85 頁。

[6] 本森, D.M., OCHEKWU, E.B. 和 TANEE, F.G. (2016) 使用牛糞和雙氧水作為修復劑的單一和組合對原油污染土壤的增強作用。應用科學與環境管理雜誌, 20(4), 第 1137-1145 頁。

[7] AKURU, U.B., ONUKWUBE, I.E., OKORO, O.I. 和 OBE, E.S. (2017) 尼日利亞實現 100% 可再生能源。可再生和可持續能源評論, 71, 第 943-953 頁。

[8] 能源百科。(2019 年, 5 月 27 日)。尼日利亞能源形勢。[在線] 可從: <https://www.energypedia.info/wiki/NigeriaEnergySituation> 獲得。[訪問 18/07/21]。

[9] AKUR, U.B. 和 OKORO, O.I. (2014) 尼日利亞的可再生能源投資: 可再生能源總體規劃審查。南部非洲能源雜誌, 25(3), 第 62-67 頁。

[10] HARJANNE, A. 和 KORHONEN, J.M. (2019) 放棄可再生能源的概念。能源政策, 127, 第 330-340 頁。

[11] CHILAKPU, K.O. (2015) 可再生能源。他們在尼日利亞的優勢、潛力和挑戰。能源技術與政策雜誌, 5(9), 第 21-24 頁

[12] COSTE, A.S. 和 ADEOTI, A.I. (2015) 氣候變化對尼日利亞玉米生產和農民適應戰略的經濟影響: 李嘉圖方法。農業科學雜誌, 7(5), 第 67 頁。

[13] 維基百科。氣候變化。[在線] 可從: https://en.wikipedia.org/w/index.php?title=Climate_change&oldid=1026669630 獲得。[訪問 18/07/21]。

[14] SHAABAN, M. 和 PETINRIN, J.O. (2014) 尼日利亞的可再生能源潛力: 滿足農村能源需求。可再生和可持續能源評論, 29, 第 72-84 頁。

[15] AGBO, E.P., EDET, C.O., MAGU, T.O., NJOK, A.O., EKPO, C.M. 和 LOUIS, H. (2021) 太陽能: 尼日利亞發電危機的靈丹妙藥。太陽神, 7(5), p.e07016。

[16] ABDULLAHI, D.、SURESH, S.、RENUKAPPA, S. 和 OLOKE, D. (2017) 八月。在尼日利亞實施太陽能的主要障礙: 批判性分析。2017 眼壓會議系列: 地球與環境科學, 83(1), 012015。

[17] ADEKUNLE, O.R., JAAFAR, A.B. 和 MUSA, M.N., (2015) 太陽能熱技術在尼日利亞可持續發展的潛力。國際工程與技術研究雜誌, 3(7), 第 2454-4698 頁。

[18] OZOEGWU, C.G., MGBEMENE, C.A. 和 OZOR, P.A. (2017) 尼日利亞太陽能整合現狀和政策。可再生和可持續能源評論, 70, 第 457-471 頁。

[19] AUDU, M.O., TERWASE, A.S. 和不列顛哥倫比亞省 ISIKWUE (2019) 尼日利亞中北部馬庫爾迪風速特徵和能源潛力調查。SN 應用科學, 1(2), 178 (2019)。

[20] ARIRIGUZO, J.C. 和 EKWE, E.B. (2019) 尼日利亞烏姆迪克風力發電前景威布爾分佈分析。機器人和計算機集成製造, 55, 第 160-163 頁。

[21] AKOREDE, M. F., O. IBRAHIM, S. A. AMUDA, A. O. OTUOZE, 和 OLUFEAGBA, B. J. (2017) 尼日利亞可再生能源發展現狀與展望。尼日利亞技術雜誌, 36(1), 第 196-212 頁。

[22] OLULEYE, A. 和 ADEYEWA, D. (2016) 根據時代臨時重新分析數據集估計的尼日利亞風能密度。當前應用科學與技術雜誌, 17(1), 第 1-17 頁。

[23] ADEDIPE, O., ABOLARIN, M.S. 和 MAMMAN, R.O. (2018) 尼日利亞陸上和海上風能潛力回顧。2018 眼壓會議系列: 材料科學與工程, 413(1), 012039。

[24] MOHAMMED, Y.S., MUSTAFA, M.W., BASHIR, N. 和 IBRAHIM, I.S. (2017) 基於自主能源和微電網技術的尼日利亞現有和推薦的可再生和可持續能源發展。可再生和可持續能源評論, 75, 第 820-838 頁。

[25] EBHOTA, W.S. 和 TABAKOV, P.Y. (2018) 小水電電氣化的地方尼日利亞社會經濟刺激項目。國際低碳技術雜誌, 13(4), 第 311-319 頁。

[26] OMEIZA, I.O.A., OJONGBEDE, H.A., THOMAS, C.T. 和 OLUFEAGBA, B.J. (2019) 2005-2014 年尼日利亞傑巴水電站渦輪發電機性能評估。尼日利亞技術雜誌, 38(1), 第 134-141 頁。

[27] OKEDU, K.E., UHUNMWANGHO, R. 和 ODJE, M. (2020) 利用尼日利亞南部克

羅斯河州小水電的潛力。可持續能源技術和評估, 37, 100617。

[28] OGUEKE, N.V., IKPAMEZIE, I.I. 和 ANYANWU, E.E. (2016) 尼日利亞富托用於發電的小型水電/光伏混合系統的潛力。國際環境能源雜誌, 37(3), 第 256-265 頁。

[29] ADEJUMOBI, I.A. 和 SHOBAYO, D.I. (2015) 小型水力發電水輪機的優化選擇——以尼日利亞西南部奧佩基河為例。尼日利亞技術雜誌, 34(3), 第 530-537 頁。

[30] NYIKA, J.、ADEDIRAN, A.A.、OLAYANJU, A.、ADESINA, O.S. 和 EDOZIUNO, F.O. (2020) 非洲生物質的潛力及其碳中和的辯論。[在線] 可從：<https://www.intechopen.com/online-first/the-potential-of-biomass-in-africa-and-the-debate-on-its-carbon-neutrality> 獲取。[訪問 18/07/21]。

[31] BEN-IWO, J.、MANOVIC, V. 和 LONGHURST, P. (2016) 尼日利亞生產運輸燃料的生物質資源和生物燃料潛力。可再生和可持續能源評論, 63, 第 172-192 頁。

[32] OLANREWAJU, F.O., ANDREWS, G.E., LI, H. 和 PHYLAKTU, H.N. (2019) 尼日利亞的生物能源潛力。化學工程交易, 74, 第 61-66 頁。

[33] IGBINOVIA, F.O. (2014) 尼日利亞可再生能源潛力概覽：前景挑戰和前進方向。能量學雜誌, 46, 第 570-579 頁。

[34] OYEDEPO, S.O. (2012) 關於尼日利亞可持續發展的能源。可再生和可持續能源評論, 16(5), 第 2583-2598 頁。

[35] SAMBO, A.S. (2009) 尼日利亞可再生能源的戰略發展。國際能源經濟學協會, 16(3), 第 15-19 頁。

[36] DIJI, C.J. (2013) 尼日利亞能源情景的批判性回顧。IOSR 電氣和電子工程雜誌, 5(3), 第 55-67 頁。

[37] OYEDEPO, S.O., BABALOLA, P.O., NWANYA, S., KILANKO, O.O., LERAMO, R.O., AWORINDE, A.K., ADEKEYE, T., OYEBANJI, J.A., ABIDAKUN, O.A. 和 AGBEREGHA, O.L. (2018) 在尼日利亞實現可持續電力供應：分散式可再生能源系統的作用。歐洲可持續發展研究雜誌, 2(4), 第 40 條。[在線] 可從：<https://www.ejosdr.com/article/towards-a-sustainable-electricity-supply-in-nigeria-the-role-of-decentralized-renewable-energy-3908> 獲得。

[38] OKEDU, K.E., UHUNMWANGHO, R. 和 WOPARA, P. (2015) 尼日利亞的可再生能源：山區和河流地區的挑戰和機遇。國際可再生能源研究雜誌 (伊爾), 5(1), 第 222-229 頁。

[39] NNAJI, C.E., UZOMA, C.C. 和 CHUKWU, J.O. (2010) 可再生能源在尼日利亞扶貧和可持續發展中的作用。大陸社會科學雜誌, 3(31), 第 31-37 頁。

[40] AKINBAMI, J.F.K. (2001) 尼日利亞的可再生能源資源和技術：現狀、前景和政策框架。全球變化的減緩和適應戰略, 6(2), 第 155-182 頁。